SAMPLE

Famous Astronomers: Gerard Kuiper

The Belt Beyond Neptune

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Free sample: Introduction & Chapter 1

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Introduction

There is a zone of ice at the edge of the solar system, a ring of frozen worlds beginning at the orbit of Neptune and stretching outward toward interstellar space, and its name is the Kuiper belt. The name attaches to a Dutch-American astronomer who died in 1973, more than a decade before the first object in that zone besides Pluto was found, and the irony of the naming is one of the threads this book follows. Gerard Peter Kuiper did not discover the belt. He proposed, in 1951, that a disc of small bodies should have formed in the outer solar system from the leftovers of planet building, and then argued, wrongly as it turned out, that most of it had since been scattered away. When the objects were finally found in the 1990s, the community reached back and gave his name to the region anyway, because the conjecture had pointed the way.

Kuiper deserves the honor for more than the conjecture. Across a forty-year career he remade the study of the solar system almost single-handedly: he found an atmosphere on Titan in 1944, the first detection of an atmosphere on any moon; he discovered Miranda around Uranus and Nereid around Neptune, the last moons to be found by one person's eyes at a telescope; he identified carbon dioxide in the atmosphere of Mars; he proved that the Martian polar caps were mostly dry ice rather than frozen water; he built the photographic atlases of the Moon that the Apollo program used to choose its landing sites; he turned a converted airliner into an infrared observatory and opened a whole spectral window on the universe; and he founded the Lunar and Planetary Laboratory at the University of Arizona, the institution that still trains a large share of the world's planetary scientists.

Colleagues called him, and historians still call him, the father of modern planetary science, and the title is not decoration. Before Kuiper, the planets were a backwater of astronomy. After him, they were a discipline.

The life behind those achievements was narrow and absolute. Born in 1905 in a Dutch village, the son of a tailor, he decided on astronomy in childhood, taught himself what his school did not offer, and never seriously considered any other career. His eyesight was legendarily sharp; he could see stars too faint for normal vision, and he used to say that the naked-eye limit for him was near magnitude seven and a half. He was a demanding man, capable of great generosity toward students and collaborators, and of public falling-out with old friends, of which the bitter split with the chemist Harold Urey over the Moon's history became the most famous. He worked constantly, considered leisure a form of inefficiency, and died of a heart attack in 1973, at sixty-eight, on vacation in Mexico City, with the lunar laboratory he had founded barely a

decade old.

The chapters that follow run on the series plan. They trace the discovery of the man himself, from a village in North Holland through Leiden at one of its golden moments, to the American observatories where he did his life's work. They explain his central scientific ideas, the observational programs behind them, and the instruments and institutions he built to pursue them. They set him in his context, the Second World War, the space race, and the birth of a new scientific profession; they follow the challenges he met and the mysteries he left open; and they carry his story past his death, to the discovery of the belt that carries his name, to the spacecraft that have flown through it, and to the laboratories and observatories still working in his shadow. The aim throughout is the series' usual one: to show what one stubborn observer contributed to knowledge, and what his successors built on the contribution.

Two notes before the story starts. First, the pronunciation: the astronomer and the belt are customarily pronounced to rhyme with viper, whatever the Dutch spelling suggests, and this book follows the custom. Second, the title father of modern planetary science is used advisedly. It is the phrase his colleagues used, the phrase embedded in the name of his field's highest honor, and the chapters ahead earn it rather than assume it, by showing what the field was before him, what he built, and what it became. The reader who wants the man and the belt can start at either end; the reader who wants the whole argument should read in order.

One more framing note. The chapters that follow use the series' standard plan, but the material arranges itself naturally into three arcs: the Dutch and Chicago years, in which the observer was formed and the early discoveries fell; the lunar and airborne years, in which the founder dealt his instruments into the space age; and the posthumous arc, in which the belt, the laboratory, and the field he founded kept producing. The reader can follow the arcs in order or take the discoveries chapter as a summary and then read backward to see how each discovery was made. Either way, the book's argument is the same: the modern study of the solar system has a founder, and this is his story.

The village boy who could see stars four times fainter than anyone else grew up to prove that the solar system's small print was worth reading, and the proof changed what astronomy is for. That, in one sentence, is the life this

book records, and the chapters that follow give it the room it earned.

Chapter 1: The Discovery of Gerard Kuiper

The making of this particular scientist began in a village called Tuitjenhorn, in

the province of North Holland, where Gerrit Pieter Kuiper was born on December 7, 1905. His father was a tailor, the household was modest, and the village was the kind of place where a bright child's horizons are set by whatever books can be reached. Kuiper reached further than most. The standard story, which he told himself in later life, is that he became fascinated by astronomy around age six or seven after receiving a small telescope, or in some accounts after simply reading about the stars, and that the fixation never loosened. Dutch schooling of the era did not offer much astronomy, so he educated himself in it while otherwise a normal pupil, and he was noticed early for an unusual asset: his eyes.

The eyesight became part of his scientific legend. As an adult he was tested and reported as able to see stars as faint as magnitude seven and a half without optical aid, several times dimmer than the typical naked-eye limit near magnitude six. The measurement is exactly the kind of claim he would have wanted to be true, but it is also the kind that fits everything else documented about his observing: he saw things at the eyepiece that others genuinely missed, and his discoveries at Uranus and Neptune were made with instruments that others had used without result. Whether the acuity was as extreme as reported or merely extraordinary, the pattern it set, that his edge was perception sharpened by preparation, ran through his whole career.

His family circumstances nearly ended the story before it started. The custom of the time would have had him leave school for a trade after the lower grades, and his father, a practical man, is said to have hoped his son would join the tailoring business. The intervention that changed the plan is usually credited to his grandmother, who argued that a boy with such marks should continue in school. He passed through the secondary schools with strong results, taught by teachers who had little astronomy but plenty of mathematics, and set his sights on the university at Leiden, one of the great centers of Dutch science.

Leiden in the 1920s was probably the best place in the world to learn astronomy. The department's faculty and associates during Kuiper's years read like a roll call of the century's astronomy: Ejnar Hertzsprung, the star-classification pioneer; Willem de Sitter, the cosmologist; Antonie Pannekoek, the astrophysicist and astronomer of the Milky Way; Jan Oort, the man who mapped the galaxy's rotation; and, in physics, Paul Ehrenfest, who taught the theoretical curriculum. Kuiper arrived in 1924, took his candidate degree in 1927, and moved directly into graduate study. His fellow students included Bart Bok and Pieter Oosterhoff, both of whom became major astronomers in their own right, and the friendships formed in that cohort lasted through their careers.

For his doctorate, Kuiper worked under Hertzsprung on binary stars, the close double systems whose orbits can be weighed and whose interactions reveal stellar masses. He took his degree in 1933 with a thesis on binary-star statistics and orbits. The subject looks, in hindsight, like an odd beginning for a planetary scientist, but it trained exactly the muscles he later used: the measurement of faint objects near bright ones, the physics of bodies orbiting under mutual gravity, and the patience of long observational programs. Binary stars also paid a professional dividend that pure astronomy rarely offered a poor Dutch graduate in the Depression: they were a field in demand in America, and America was where the telescopes were.

The crossing came in 1933, with a fellowship to the Lick Observatory in California, to work under Robert Grant Aitken, the dean of American double-star astronomy. From a Dutch village to the mountains above San Jose Bay in a single decade, by way of self-teaching, one supportive grandmother, and extraordinary eyes: that is the discovery of Gerard Kuiper, and every later chapter of this book runs on the momentum it generated.

One more look at the formative stretch is worthwhile, since the pattern of the whole career was fixed before he ever left Holland. The village context matters: Tuitjenhorn in North Holland was farm and trade country, a tailor's son had no obvious road to a university, and the Dutch school system of the era was designed to move most children into work early. That Kuiper escaped the default track at all required the intervention already mentioned, the grandmother who insisted the bright boy continue, and the Dutch tradition of rigorous public examination that let talent prove itself. He proved it in the state examinations, learned English and the other languages of science on his own, and arrived at Leiden at nineteen with a preparation in mathematics that let him stand in the same lectures as students from elite schools.

At Leiden the intellectual weather did the rest. He arrived in 1924, took the candidate degree in 1927, and went straight into graduate work, a fast professional start by any standard. The physics culture of the university in those

years, with Ehrenfest's famous colloquium and the astronomy of Hertzsprung and de Sitter, treated students as junior colleagues and hard problems as ordinary conversation, and Kuiper, intense, prepared, and sure of his direction, fitted the culture exactly. He made lifelong friends among his classmates, Bok and Oosterhoff prominent among them, and the Dutch network he built as a student remained one of his professional assets for the rest of his life. The thesis on binary stars, completed under Hertzsprung in 1933, made him at twenty-eight a trained stellar astronomer with a specialty, a reputation, and,

crucially, a field in demand in the United States, where the large telescopes and the double-star programs waited.

A last word on the childhood, because the series' biographies keep finding the pattern: the childhood fascination that survives contact with professional training becomes a career, and the career becomes a discipline. The boy with the extraordinary eyes and the village library book grew into the man who could see moons nobody else could find, and the connection is not metaphorical. The acuity was a gift of birth; the use of the acuity was a decision, made early, in a village with no telescopes, and held for sixty years. The chapters ahead are the record of what the decision produced, and they begin, as the career did, with the journey from Tuitjenhorn to Leiden.

One trip detail completes the crossing story: the fellowship that brought him from Leiden to California in 1933 was the standard currency of the era's scientific migration, and the young astronomer who arrived at Lick Observatory, a mountain station above the Santa Clara valley, found the largest American telescopes, a vigorous double-star program under Robert Aitken, and a professional culture that put the observer first. The fit was immediate and permanent. Within two years he had moved to Harvard, within four he had chosen Chicago over Java, and by the end of the decade the Dutch student was an American scientist with a career under way, the whole crossing completed by the age of thirty-two.

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